

Identifying Interaction Design and Functional Layout

Practical implementation of interaction design and functional layouts

Case study: **Personal App — Habits & Calendar Modules**

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Tools: Flutter 3.22+ · Dart 3.7+ · Provider · SQLite

1. Aim

To identify user requirements for a habit tracking and calendar management system, design interaction prototypes, and implement them with consistent placement, wording, and colour so the user can build an accurate mental model — demonstrated through the **Habits and Calendar modules** of Personal App.

2. Objectives

1. Understand the principles of interaction design and functional layout.
2. Translate user requirements into working design prototypes for Habits and Calendar (Activity 1).
3. Apply consistency rules across every screen in both modules (Activity 2).
4. Evaluate whether the resulting interface supports an accurate, predictable mental model.

3. Theory

3.1 Interaction Design

Interaction design focuses on the sequence of screens and states a user moves through to complete a task. In Habits, this includes the flow from the dashboard to the weekly checklist, monthly stats, and habit creation. In Calendar, this includes navigating months, viewing day details, and creating or editing events.

3.2 Functional Layout

Both modules use a consistent functional layout: FAB at bottom-right for creation, search bar at the top, and a scrollable content area. The Habits module uses a combination of horizontal scrollable cards, weekly rows, and monthly grids. The Calendar module uses a month grid with touch targets for days.

3.3 Consistency Principles

- **Placement** — FAB, search bar, and navigation elements are in identical positions across both modules.
- **Wording** — “Save”, “Cancel”, “Delete”, “Discard” use identical labels.
- **Colour** — Blue = primary action, red = destructive, amber = warning, green = success.

3.4 Mental Model

When placement, wording, and colour stay consistent between Habits and Calendar, a user who learns one module can predict how the other will behave, reducing learning time.

4. Software / Tools Used

Tool	Purpose
Flutter 3.22+	Cross-platform UI framework (Material 3)
Dart 3.7+	Programming language with sound null safety
Provider	State management (HabitsProvider, CalendarProvider)
SQLite (sqflite)	Local database (habits + logs, events)
flutter_local_notifications + timezone	Habit reminders + event notifications
shared_preferences	User preferences
intl	Date/time formatting

5. Activity 1 — Design and Prototype the Interaction

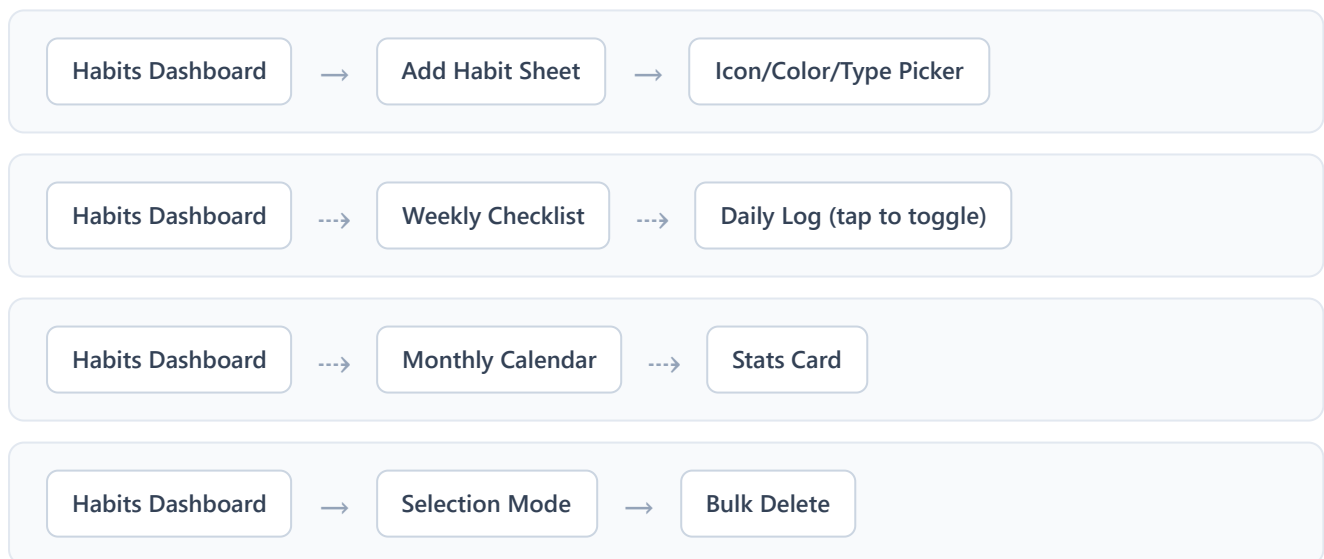
Module 1 Habits

5.1 User requirements — Habits

#	User requirement	Screen / interaction
1	Create habits with a name, icon, and colour.	Add Habit sheet — icon picker (55 icons), color picker (12 presets + custom)

#	User requirement	Screen / interaction
2	Choose a tracking type for each habit.	Type selector — Yes/No, Count+Target, Free Count
3	Log daily progress with a single tap.	Weekly checklist — tap to toggle, +/- for count types
4	View weekly progress at a glance.	Weekly checklist row — 7-day horizontal with nav arrows
5	View monthly completion in a calendar grid.	Monthly log calendar — color-coded completion cells
6	Track habit streaks.	Stats card — current streak + best streak
7	Search habits by name.	Search bar — real-time filtering
8	Reorder habits by dragging.	Cards — drag to reorder within horizontal scroll
9	Set a daily reminder.	Add/Edit habit — time picker + repeating notification
10	Bulk delete via multi-select.	Selection Mode — long-press, select-all, delete
11	See today's overall progress.	Today Progress Card — X of Y done + progress bar
12	Quick suggestions for common habits.	ActionChips — Read, Meditate, Walk, Drink Water

Screen flow — Habits



Views — Habits

Habits Module — View Inventory

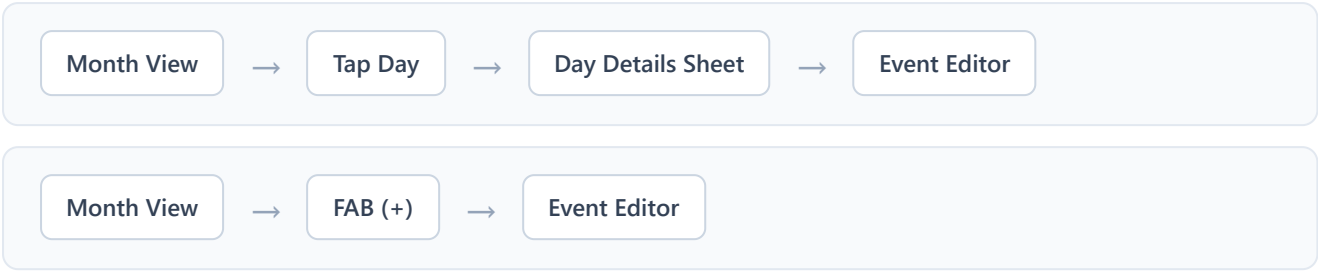
Category	Views
Dashboard	Today Progress Card, Horizontal Habit Cards, Weekly Checklist, Monthly Calendar, Stats Card
Creation	Add Habit Sheet, Edit Habit Sheet
Pickers	Icon Picker (9 categories, 55 icons), Color Picker (12 presets + custom), Type Selector
Search	Search with results, Search no results
Actions	Selection Mode, Delete Confirmation dialog

Module 2 Calendar

5.2 User requirements — Calendar

#	User requirement	Screen / interaction
1	View a monthly calendar with navigation.	Month grid — swipe/arrows for prev/next month
2	Create events with title, date, time, category.	Event Editor sheet — form + category dropdown
3	See event indicators on calendar days.	Month grid — colored dots on days with events
4	Filter events by category.	Filter chips — General, Work, Personal, Urgent, All
5	Search events by title or notes.	Search bar — real-time filtering
6	Set event recurrence.	Recurrence dropdown — None/Daily/Weekly/Monthly + end date
7	Receive event notifications.	Auto-scheduled notification at event time
8	Edit or delete events from day view.	Day Details sheet — Edit/Delete buttons
9	Today is always highlighted.	Month grid — circle highlight on current date
10	Confirm before losing unsaved changes.	Discard Changes dialog

Screen flow — Calendar



Views — Calendar

Calendar Module — View Inventory

Category	Views
Grid	Month Grid (with events), Month Grid (no events)
Day	Day Details (with events), Day Details (empty)
Events	Event Editor (create), Event Editor (edit)
Search	Search with results, Search no results
Dialogs	Discard Changes dialog, Delete confirmation

6. Activity 2 — Apply and Verify Consistency

Consistency rules were applied once and reused across both the Habits and Calendar modules via the shared Material 3 design system.

6.1 Consistency rules applied

Button Placement

FAB is always bottom-right in both Habits and Calendar. Delete/destructive actions always show confirmation dialogs. Save is top-right in forms. Cancel is left, Confirm is right in dialogs.

Label Wording

'Save', 'Cancel', 'Delete', 'Discard', 'Retry' never change. Confirmation dialogs echo the verb: 'Delete habit?', 'Delete event?'.

Colour Scheme

-  **Blue** — primary action, FAB, active tab
-  **Red** — destructive (delete, discard)
-  **Green** — success, completion
-  **Amber** — warning, medium priority

Navigation & Icons

Search bar is always top-left. Bottom NavigationBar persists with 6 tabs. Icons use the same symbol set (trash, +, back, search). IndexedStack preserves module state when switching tabs.

6.2 Consistency Matrix

Cross-Module Consistency Matrix — Habits & Calendar

Rule	Habits	Calendar
FAB bottom-right	✓	✓
Search top-left	✓	✓
Blue = primary action	✓	✓
Red = destructive	✓	✓
Confirm before delete	✓	✓
Same wording for actions	✓	✓
Error/Loading/Empty states	✓	✓
Shared theme applies	✓	✓

7. Output / Result

The Habits and Calendar modules are implemented within a single Flutter application sharing the Material 3 design system. The Habits module supports three tracking types (Yes/No, Count+Target, Free Count) with weekly and monthly views, streak tracking, and search. The Calendar module provides a monthly grid with event management, category filtering, recurrence, and notifications. Both modules share the same FAB placement, search pattern, dialog style, and colour semantics.

7.1 Self-evaluation checklist

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- ✓ **Primary action in same position?** — FAB bottom-right in both modules
-
- ✓ **Same action = same word?** — Save/Cancel/Delete/Discard identical
-
- ✓ **Each colour = one meaning?** — Blue=action, Red=destructive, Green=success, Amber=warning
-
- ✓ **Predict unseen screen?** — Calendar reuses FAB, search, dialog patterns from Habits
-
- ✓ **Loading/error states handled?** — Both have error states with retry
-
- ✓ **Destructive actions protected?** — Always requires confirmation dialog
-
- ✓ **Empty/populated states?** — Both have empty states with CTAs
-
- ✓ **Navigation persists?** — 6-tab IndexedStack
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Annexure A — Selected UI/UX Ideas Integrated

1. Streak Tracking for Habit Motivation

Module: Habits | Current streak and best streak calculated from consecutive daily completions. Displayed in the Stats Card alongside monthly completion percentage. Drives daily engagement through the “don’t break the chain” effect.

2. Monthly Calendar with Completion Colours

Module: Habits | The monthly log calendar colour-codes each cell based on completion status. Full completion = green, partial = amber, none = gray. Users see their monthly pattern at a glance.

3. Three Tracking Types for Flexibility

Module: Habits | Users choose from Yes/No (simple toggle), Count+Target (e.g., 8 glasses of water), or Free Count (unlimited tracking). Each type has a dedicated UI with +/- buttons or circular progress.

4. Category-Based Event Filtering

Module: Calendar | Events are tagged with categories (General, Work, Personal, Urgent) and colour-coded. Users filter events by category using chip toggles. The filter state persists across month navigation.

5. Recurring Event Support

Module: Calendar | Events support daily, weekly, and monthly recurrence with an optional end date. Notifications are rescheduled automatically based on recurrence rules.

Annexure B — Screen Wireframes (Habits & Calendar)

Six representative screens from the Habits and Calendar modules.

